ANANYA SRIDHAR

AI Engineer — Generative AI, LLMs & Agentic Systems

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GitHub: github.com/darayavahoush · Portfolio: darayavahoush.github.io · LinkedIn: linkedin.com/in/ananya-sridhar-ai · Vaaksudhi Live · Vaakify Live

Languages: English, Hindi, Tamil

PROFESSIONAL SUMMARY

AI Engineer specializing in production Generative AI, multi-agent workflows (LangGraph), and deterministic RAG systems. Proven track record deploying enterprise-grade LLM applications serving 10,000+ financial advisors (Nivesh AI) and clinical multi-agent speech intelligence platforms across 8 Indian languages (Manas Learning). Skilled in low-latency REST APIs (FastAPI, WebSockets), vector retrieval (ChromaDB, pgvector), prompt engineering, and cloud deployments on Azure with automated CI/CD.

TECHNICAL SKILLS

- **Languages:** Python (Proficient), C++, SQL, TypeScript, JavaScript, Bash
- **Generative AI & Agentic Systems:** LangGraph, LangChain, Multi-Agent Architecture, RAG (Retrieval-Augmented Generation), Prompt Engineering, Vector Databases (ChromaDB, pgvector), LLM Evaluation, Function Calling
- **Deep Learning & Frameworks:** PyTorch, Hugging Face (Transformers, PEFT/LoRA, Accelerate), Scikit-Learn, Reinforcement Learning (Q-Learning, DRL)
- **Speech & Multimodal AI:** Faster-Whisper, OpenAI Whisper, Vosk, Kokoro ONNX, Audio Signal Processing, Librosa, Epitran (G2P), OpenCV, MediaPipe
- **Cloud, Backend & DevOps:** FastAPI, REST APIs, Microservices, WebSockets, Azure Cloud, Docker, PostgreSQL, Redis, GitHub Actions CI/CD, Linux, Uvicorn, System Design

PROFESSIONAL EXPERIENCE

AI/ML ENGINEER INTERN (AGENTIC AI & SPEECH SYSTEMS) | MANAS LEARNING

Mar 2026 – Present | Remote / Mumbai, India

- **LangGraph Multi-Agent Architecture:** Architected an autonomous diagnostic agentic workflow chaining Faster-Whisper, Vosk, and Kokoro TTS with phoneme-level acoustic verification across 8 Indian languages.
- **Diagnostic State Machines:** Implemented LangGraph state machines coordinating 4 diagnostic assessment modules, evaluating place/manner/voicing speech errors with sub-300ms latency.
- **Automated Clinician Reporting:** Designed automated structured report generation converting raw acoustic evaluation into diagnostic clinical summaries for speech therapists.
- **Adaptive Reinforcement Learning:** Formulated a Q-learning difficulty controller that dynamically personalizes practice sessions based on rolling acoustic accuracy and response latency.
- **Production API & Microservice Delivery:** Engineered scalable REST APIs and WebSocket endpoints using FastAPI and PostgreSQL, implementing database connection pooling and Alembic migrations.
- **Cloud Deployment & Security:** Containerized backend services with Docker, established GitHub Actions CI/CD pipelines, and integrated COPPA-compliant parental consent and secure JWT authentication.

AI/ML ENGINEER INTERN (ENTERPRISE AI & RAG) | NIVESH AI

Jun 2026 – Present | Mumbai, India

- **Enterprise RAG Pipeline on Azure:** Developed a zero-hallucination RAG summarization engine deployed on Azure Cloud, processing advisor-client audio and portfolio documents for 10,000+ mutual fund distributors (MFDs).
- **Multilingual Speech-to-Text Optimization:** Integrated and fine-tuned domain-specific STT models (Whisper, Sarvam AI) with spectral denoising (`noisereduce`), boosting transcription fidelity on mixed Indian English and Hindi audio.
- **Structured Knowledge Extraction:** Automated ingestion of complex financial portfolios and transcripts into ChromaDB vector stores using `pdfplumber`, `python-docx`, and embedding models.
- **Business & Operational Impact:** Slashed advisor documentation turnaround time by 80%, automating manual note-taking and compliance tracking across tens of thousands of client interactions.
- **Cloud Reliability & Monitoring:** Containerized backend services using Docker on Azure, set up asynchronous background task queues, and implemented health checks and structured logging.

FEATURED PROJECTS

VAAKSUDHI — PEDIATRIC SPEECH & DIAGNOSTIC MULTI-AGENT PLATFORM

Live Platform · GitHub

Python, LangGraph, Faster-Whisper, Vosk, Kokoro ONNX, Librosa, Q-Learning, FastAPI, React

- Architected an autonomous multi-agent clinical diagnostic pipeline coordinating 4 clinical assessment modules across 8 Indian languages.
- Implemented LangGraph state machines evaluating speech errors with sub-300ms phoneme alignment, auto-generating structured clinician diagnostic reports.
- Live in active clinical use with automated error taxonomy categorization and personalized practice games.

NYAYAI & AI BAIL RECKONER — INDIAN LEGAL-REASONING LLM & RAG SYSTEM

GitHub: NyayAI · GitHub: Bail Project

PyTorch, Hugging Face Transformers, PEFT/LoRA, ChromaDB, FastAPI, React

- Created a specialized legal AI assistant predicting bail hearing outcomes and citing judicial precedents across the Indian Penal Code (IPC) and Bharatiya Nyaya Sanhita (BNS).
- Implemented a dense RAG pipeline using ChromaDB vector embeddings to ground predictions strictly in verified Supreme Court rulings, eliminating hallucinations.
- Accelerated legal precedent retrieval from hours to seconds with verifiable case citations.

AEGIS — MULTI-AGENT SIMULATION & DECISION PLANNING

Live Simulation Console · GitHub

Python, NetworkX, NumPy, A Pathfinding, MCTS, Reinforcement Learning, FastAPI, React*

- Developed a multi-agent simulation coordinating autonomous drone swarms and rescue teams under time, weather, and battery constraints.
- Engineered hazard-aware routing and Monte Carlo Tree Search (MCTS) for dynamic task assignment and real-time obstacle replanning.
- Built a WebSocket-driven command console deployed live on Render streaming agent trajectories and mission metrics.

NIRVANA (FIREBOT) — AUTONOMOUS AGENT & 3D SIMULATION

Live Render Console · GitHub

Python, MuJoCo Physics, PyTorch, Gymnasium, FastAPI, WebSockets, Three.js

- Built an autonomous agent operating in a MuJoCo 3D simulator across procedural multi-room environments with particle fire dynamics.
- Implemented deep reinforcement learning with multimodal sensor fusion for autonomous obstacle navigation and fire source localization in <45s.
- Deployed a real-time web telemetry console on Render streaming robot telemetry and event logs via FastAPI and WebSockets.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering

SASTRA Deemed University, India

Expected Graduation: 2027

CERTIFICATIONS & RECOGNITIONS

- **SWAYAM – IIT Madras:** Comprehensive Electronics and Embedded Systems *(In-Person Practical Program)*
- **IIoT Certification:** Ministry of Heavy Industries (MHI) Program *(Project-Based Evaluation)*